Paul (Yi-Chia) Chen

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ECE Ph.D. student at USC researching compiler-directed optimization for energy-efficient machine learning accelerators and silicon prototyping.

EDUCATION

University of Southern California

Los Angeles, California

- **Ph.D., Electrical and Computer Engineering** | Aug 2024 - Present
  Advisor: Prof. Christopher Torng, ACORN Research Group
- **M.S., Computer Engineering** | Aug 2023 - May 2024
- **B.S., Electrical and Computer Engineering** | Aug 2019 - May 2023
  W.V.T. Rusch Honors Program Research Award

SELECTED RESEARCH & SILICON PROJECTS

PowerFlow-DNN Accelerator

USC ACORN Lab | TSMC 40 nm | Dec 2025 - Present

- Built compiler-directed DVFS and power-gating schedules for edge AI inference, reducing energy by up to 48% under application-level performance constraints.
- Completed synthesis, place-and-route, static timing analysis, SDF-annotated gate-level simulation, and post-layout power characterization for the ISLPED 2026 evaluation.

Chipstitch: Accelerated Silicon Prototyping

USC ACORN Lab | ICCAD 2026, forthcoming November 2026

- Contributed clock-tree synthesis and clocking strategies for heterogeneous block integration across Intel 16 nm and TSMC 28 nm HPC+.

Multi-Precision LLM Accelerator

USC Advanced Chip Design and Tapeout (EE 599) | Aug 2024 - Dec 2024

- Designed a BERT-layer accelerator with configurable 4-, 8-, and 16-bit QKV datapaths; taped out in Intel 16 nm.
- Led a six-person team through RTL-to-GDS implementation and SDF-annotated gate-level simulation; achieved timing closure at 1 GHz.

GNN Accelerator on Xilinx Versal ACAP

USC FPGA/Parallel Computing Lab | Jun 2022 - Jun 2023

- Co-designed GNN inference across programmable logic and AI Engines, achieving a 162x speedup over the CPU baseline; published at FPL 2023.

TECHNICAL SKILLS

Design & implementation: SystemVerilog/Verilog, RTL verification, CDC, FPGA, synthesis, physical design, STA, CTS, power planning, RTL-to-GDS.
Tools: Cadence Genus/Innovus; Synopsys VCS/PrimeTime/Verdi; Siemens Calibre.
Programming: Python, C++, CUDA, PyTorch.
Processes: TSMC 40 nm LP, TSMC 28 nm HPC+, Intel 16 nm.

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Publications, industry experience, and teaching

PUBLICATIONS

ICCAD 2026 | Forthcoming, November 2026

IEEE/ACM International Conference on Computer-Aided Design

- J. Kim, S. Yarzada, **P. YC. Chen**, Y. Li, W. Zhu, and C. Torng. "Chipstitch: A Scalable and Cost-Efficient Platform for Accelerated Silicon Prototyping."

ISLPED 2026 | Best Paper Candidate

IEEE/ACM International Symposium on Low Power Electronics and Design | August 2026

- **P. YC. Chen**, J. Kim, W. Zhu, Y. Li, S. Huang, C. Weng, and C. Torng. "PowerFlow-DNN: Compiler-Directed Fine-Grained Power Orchestration for End-to-End Edge AI Inference."
- Code and paper artifact

FPL 2023 | Co-first author

IEEE International Conference on Field-Programmable Logic and Applications | September 2023

- **P. Chen***, P. Manjunath*, S. Wijeratne, B. Zhang, and V. Prasanna. "Exploiting On-Chip Heterogeneity of Versal Architecture for GNN Inference Acceleration." (* Equal contribution)
- Paper: arxiv.org/abs/2308.02749

INDUSTRY EXPERIENCE

Machine Learning Engineer Intern

EMD Electronics (Merck KGaA) | Carlsbad, CA | Jul 2023 - Jul 2024

- Developed linear, XGBoost, and random-forest models to identify semiconductor process features associated with yield changes.
- Optimized distributed data-processing and ML workflows, and implemented automated retraining and model-monitoring pipelines on Palantir Foundry.

TEACHING

Teaching Assistant, Advanced Chip Design and Tapeout (EE 599)

USC | Fall 2024 and Fall 2025

- Mentored 30 students across six teams through RTL-to-GDS implementation and verification, producing six designs integrated into an Intel 16 nm tapeout.

Course Lab Mentor, Introduction to Digital Circuits (EE 354)

USC | Spring 2022

- Guided FPGA-based digital circuit labs and held office hours on RTL development, simulation, and hardware debugging.

Course Grader, Introduction to C++ (EE 155)

USC | Fall 2021

- Graded C++ assignments, labs, and exams.

ENTREPRENEURSHIP

Regional I-Corps, West Hub | 2025
Completed ZAP and BOOM programs.